

Firmenich

GHS Safety Data Sheet



According to Regulation ST/SG/AC.10/30/Rev.8 UN GHS

This Safety Data Sheet cancels and replaces all preceding SDS for this product.

1 IDENTIFICATION OF THE SUBSTANCE/PREPARATION AND OF THE COMPANY/UNDERTAKING

1.1 Product identifier

LEMON LAND

© Firmenich product

1.2 Relevant identified uses of the substance or mixture and uses advised against

Fragrance ingredient for use in consumer products.
Not for personal use in this form or concentration.
Intended to be used in the manufacture of products for consumers

1.3 Details of the supplier of the safety data sheet

Firmenich Asia Private Ltd
No 10 Tuas West Road
Singapore 638377
SINGAPORE
Tel: +(65) 6347 2888

GRS.Fragrance.FIRSIN@firmenich.com

1.4 Emergency telephone number

+65 3158 1074 NCEC - Available 24/7

2 HAZARDS IDENTIFICATION

2.1 Classification of the substance or mixture

2.1.1 Classification according to Regulation ST/SG/AC.10/30/Rev.8 [UN/GHS]

Flammable Liquid - Cat. 4	H227
Skin Irritation - Cat. 2	H315
Skin Sensitization - Cat. 1	H317
Environmental Hazard (acute) - Cat. 2	H401
Environmental Hazard (chronic) - Cat. 2	H411

2.1.2 Additional information

Full text of listed statements : See section 16

2.2 Label elements

Hazard pictograms:

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Signal Word:

Warning

Hazard Statements:

H315	Causes skin irritation.
H317	May cause an allergic skin reaction.
H227	Combustible liquid
H411	Toxic to aquatic life with long lasting effects.

Precautionary Statements:

P210	Keep away from heat/sparks/open flames/hot surfaces. No smoking.
P264	Wash thoroughly after handling.
P273	Avoid release to the environment.
P302+P352	IF ON SKIN: Wash with plenty of soap and water.
P333+P313	If skin irritation or rash occurs: Get medical advice/attention.
P403+P235	Store in a well-ventilated place. Keep cool.

2.3 Other hazards

No data available at this time.

3 COMPOSITION/INFORMATION ON INGREDIENTS

Mixture of aromatic substances.

Contains :

$\geq 10.0 < 15.0\%$
Ethanone, 1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-Tetramethyl-2-Naphthalenyl)-
N° CAS : 0054464-57-2
N° ELINCS: --
N° EU REACH: 01-2119489989-04

GHS Classification:

Skin Sensitization - Cat. 1B [H317]
Skin Irritation - Cat. 2 [H315]
Environmental Hazard (chronic) - Cat. 1 [H410]
Environmental Hazard (acute) - Cat. 2 [H401]

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$\geq 2.5 < 5.0\%$

Cyclopentaneacetic Acid, 3-Oxo-2-Pentyl-, Methyl Ester

N° CAS : 0024851-98-7

N° EINECS: 246-495-9

N° EU REACH: 01-2119493355-31

Environmental Hazard (acute) - Cat. 2 [H401]

$\geq 1.0 < 2.5\%$

D-Limonene

N° CAS : 0005989-27-5

N° EINECS: 227-813-5

Aspiration Hazard - Cat. 1 [H304]

Skin Sensitization - Cat. 1B [H317]

Skin Irritation - Cat. 2 [H315]

Environmental Hazard (acute) - Cat. 1 [H400]

Environmental Hazard (chronic) - Cat. 3 [H412]

Flammable Liquid - Cat. 3 [H226]

$\geq 1.0 < 2.5\%$

Linalyl Acetate

N° CAS : 0000115-95-7

N° EINECS: 204-116-4

N° EU REACH: 01-2119454789-19

Skin Sensitization - Cat. 1B [H317]

Skin Irritation - Cat. 2 [H315]

Eye Irritation - Cat.2B [H320]

Environmental Hazard (acute) - Cat. 3 [H402]

Flammable Liquid - Cat. 4 [H227]

$\geq 0.5 < 1.0\%$

Citral

N° CAS : 0005392-40-5

N° EINECS: 226-394-6

N° EU REACH: 01-2119462829-23

Skin Sensitization - Cat. 1 [H317]

Skin Irritation - Cat. 2 [H315]

Eye Irritation - Cat.2A [H319]

Environmental Hazard (acute) - Cat. 2 [H401]

$\geq 0.5 < 1.0\%$

Ethyl Linalool

N° CAS : 0010339-55-6

N° EINECS: 233-732-6

N° EU REACH: 01-2119969272-32

Skin Sensitization - Cat. 1B [H317]

Eye Irritation - Cat.2A [H319]

Skin Irritation - Cat. 3 [H316]

Environmental Hazard (acute) - Cat. 3 [H402]

Acute Toxicity (oral) - Cat. 5 [H303]

Flammable Liquid - Cat. 4 [H227]

$\geq 0.1 < 0.5\%$

Citronellol

N° CAS : 0000106-22-9

N° EINECS: 203-375-0

N° EU REACH: 01-2119453995-23

Skin Sensitization - Cat. 1B [H317]

Skin Irritation - Cat. 2 [H315]

Eye Irritation - Cat.2A [H319]

Environmental Hazard (acute) - Cat. 2 [H401]

Acute Toxicity (oral) - Cat. 5 [H303]

Acute Toxicity (dermal) - Cat. 5 [H313]

$\geq 0.1 < 0.5\%$

Octanal, 2-(Phenylmethylene)-

N° CAS : 0000101-86-0

N° ELINCS: --

N° EU REACH: 01-2119533092-50

Skin Sensitization - Cat. 1B [H317]

Skin Irritation - Cat. 3 [H316]

Environmental Hazard (acute) - Cat. 1 [H400]

Environmental Hazard (chronic) - Cat. 2 [H411]

Acute Toxicity (oral) - Cat. 5 [H303]

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$\geq 0.1 < 0.5\%$

Beta-Pinene

N° CAS : 0000127-91-3

N° EINECS: 242-060-2

Aspiration Hazard - Cat. 1 [H304]

Skin Sensitization - Cat. 1B [H317]

Skin Irritation - Cat. 2 [H315]

Environmental Hazard (acute) - Cat. 1 [H400]

Environmental Hazard (chronic) - Cat. 1 [H410]

Flammable Liquid - Cat. 3 [H226]

$\geq 0.1 < 0.5\%$

Octanal, 7-Hydroxy-3,7-Dimethyl-

N° CAS : 0000107-75-5

N° EINECS: 203-518-7

Skin Sensitization - Cat. 1B [H317]

Eye Irritation - Cat.2A [H319]

Environmental Hazard (acute) - Cat. 3 [H402]

$\geq 0.1 < 0.5\%$

Alpha-Cedrene

N° CAS : 0000469-61-4

N° EINECS: 207-418-4

Aspiration Hazard - Cat. 1 [H304]

Skin Irritation - Cat. 2 [H315]

Environmental Hazard (acute) - Cat. 1 [H400]

Environmental Hazard (chronic) - Cat. 1 [H410]

$\geq 0.1 < 0.5\%$

Alpha-Pinene

N° CAS : 0000080-56-8

N° EINECS: 232-077-3

N° EU REACH: 01-2119979519-16

Aspiration Hazard - Cat. 1 [H304]

Skin Sensitization - Cat. 1B [H317]

Acute Toxicity (oral) - Cat. 4 [H302]

Skin Irritation - Cat. 2 [H315]

Environmental Hazard (acute) - Cat. 1 [H400]

Environmental Hazard (chronic) - Cat. 1 [H410]

Flammable Liquid - Cat. 3 [H226]

$\geq 0.1 < 0.5\%$

1,4-Cyclohexadiene, 1-Methyl-4-(1-Methylethyl)-

N° CAS : 0000099-85-4

N° EINECS: 202-794-6

N° EU REACH: 01-2120780478-40

Reproductive Toxicity - Cat. 2 [H361]

Aspiration Hazard - Cat. 1 [H304]

Environmental Hazard (chronic) - Cat. 2 [H411]

Environmental Hazard (acute) - Cat. 2 [H401]

Acute Toxicity (oral) - Cat. 5 [H303]

Flammable Liquid - Cat. 3 [H226]

$\geq 0.1 < 0.5\%$

Benzene, 1-Methyl-4-(1-Methylethyl)-

N° CAS : 0000099-87-6

N° EINECS: 202-796-7

N° EU REACH: 01-2120807345-59

Reproductive Toxicity - Cat. 2 [H361]

Acute Toxicity (inhalation) - Cat. 3 [H331]

Aspiration Hazard - Cat. 1 [H304]

Environmental Hazard (chronic) - Cat. 2 [H411]

Environmental Hazard (acute) - Cat. 2 [H401]

Acute Toxicity (oral) - Cat. 5 [H303]

Flammable Liquid - Cat. 3 [H226]

$\geq 0.1 < 0.5\%$

Linalool

N° CAS : 0000078-70-6

N° EINECS: 201-134-4

N° EU REACH: 01-2119474016-42

Skin Sensitization - Cat. 1B [H317]

Skin Irritation - Cat. 2 [H315]

Eye Irritation - Cat.2A [H319]

Environmental Hazard (acute) - Cat. 3 [H402]

Acute Toxicity (oral) - Cat. 5 [H303]

Flammable Liquid - Cat. 4 [H227]

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$\geq 0.1 < 0.5\%$

Caryophyllene

N° CAS : 0000087-44-5

N° EINECS: 201-746-1

N° EU REACH: 01-2120745237-53

Aspiration Hazard - Cat. 1 [H304]

Skin Sensitization - Cat. 1B [H317]

$\geq 0.1 < 0.5\%$

2h-1-Benzopyran-6-ol, 3,4-Dihydro-2,5,7,8-

Tetramethyl-2-(4,8,12-Trimethyltridecyl)-

N° CAS : 0010191-41-0

N° EINECS: 200-412-2

Skin Sensitization - Cat. 1B [H317]

4 FIRST-AID MEASURES

4.1 Description of first aid measures

General information:

As in all cases of potential poisoning, Obtain medical advice immediately.

In case of eye contact:

Irrigate copiously with water for at least 10 minutes. Obtain medical advice if any irritation or evidence of tissue damage persists.

In case of inhalation:

Remove the individual to fresh air and keep at rest. Obtain medical advice immediately.

In case of skin contact:

Remove contaminated clothes. Wash skin with large volumes of water.

If irritation persists, or any sign of tissue damage is apparent, obtain medical advice immediately.

In case of ingestion:

Rinse mouth with water. Obtain medical advice immediately.

4.2 Most important symptoms and effects, both acute and delayed

No information available on the product itself.

4.3 Indication of immediate medical attention and special treatment needed

None known.

5 FIRE-FIGHTING MEASURES

5.1 Extinguishing media

In the event of fire, adequate extinguishers should be used. Avoid inhalation of smoke and fumes. In case of insufficient ventilation, wear suitable respiratory equipment.

5.2 Special hazard arising from the substance or mixture

No specific hazard known.

5.3 Advice for fire-fighters

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No specific advice.

6 ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

For non-emergency personnel:

Adequate protective gloves should be worn when handling spillages. No smoking. Avoid naked flames or other potential sources of ignition (eg. electrical equipment).

Avoid skin contamination and inhalation of vapour.

Individual washing routines should be followed after any potential contact.

Ensure adequate ventilation in working areas following accidental releases.

For emergency personnel:

Apply the same recommendations as section 6.1

6.2 Environmental precautions

Do not discharge directly into drains, air, into soil or into the aquatic environment.

6.3 Methods and material for containment and cleaning up

For containment:

Any absorbent used for cleaning up spillage should be disposed promptly, accordingly to local regulation.

Gross spillages should be contained by any means, and disposal of this should be in accordance with Government Regulations.

For cleaning-up:

Spillages should be disposed of in accordance with Governmental Regulations.

7 HANDLING AND STORAGE

7.1 Precautions for safe handling

Avoid contact with skin and eyes.

Wear adequate protective gloves protection and eye/face protection.

No smoking. Avoid any sources of ignition.

Avoid exposing to high temperature during processing.

Do not ingest or apply to the skin as such. Good personal washing routines should be followed.

Maintain adequate local and general ventilation where product is handled.

7.1.1 Protective measures

Keep strict control of dust accumulation to a minimum. Maintain adequate local and general ventilation where product is handled. Avoid any sources of ignition.

7.1.2 Advice on general occupational hygiene

Good personal washing routines should be followed.

7.2 Conditions for safe storage, including any incompatibilities

It is good general practice to store in closed, preferably full, containers away from heat sources, and protected from extremes of temperature. Do not re-use the empty container.

Respect general rules for compatibility storage.

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7.3 Specific end use(s)

Not available at this time.

8 EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 Control parameters

Contains no substances with occupational exposure limit values

Derived No Effect Level (DNEL)

No data available

Predicted No Effect Concentration (PNEC)

No data available

8.2 Exposure controls

Avoid exposing to high temperature during processing.

Maintain adequate local and general ventilation where product is handled.

8.2.1 Appropriate engineering controls

Maintain adequate local and general ventilation where product is handled and dispensed.

8.2.2 Environmental exposure controls

Not available at this time. Minimize release to the environment.

8.3 Personal protection

Respiratory protection: Where ventilation (LEV) is inadequate to prevent any inhalation of substances, use adequate respiratory apparatus.

Hand protection: Adequate Protective Gloves should be worn.

Eye protection: Adequate safety glasses should be used.

Skin protection: Wear protective clothing, overall if necessary to limit the odour contamination of personal clothing. Individual washing routines should be followed after any potential contact.

9 PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

Appearance :

Colour :

Odour :

LIQUID

COLOURLESS TO PALE YELLOW

Characteristic strong odour according to the commercial description of the substance.

pH :

Melting point/range (°C) :

Initial boiling point/range (°C) :

Flash point (closed cup) :

Evaporation rate :

Flammability :

Not available

Not available

Not available

= 77°C

Not available

Flammable

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Explosive properties (St class) :	N/A
Vapour pressure (Pa) :	Not applicable
Vapour density :	Not available
Relative density (d 20/20) :	0.994 - 1.004
Water solubility (20°C) :	Not applicable
Partition coef. (n-octanol/water) :	Not applicable
Auto-ignition temperature (°C):	Not available
Decomposition temperature :	Not available
Viscosity :	Not available
Explosive properties :	Not available
Oxidizing properties :	Not available

9.2 Other safety information

None

10 STABILITY AND REACTIVITY

10.1 Reactivity

No reaction known with water.

10.2 Chemical stability

Good stability at standard temperature.

10.3 Possibility of hazardous reactions

Not known.

10.4 Conditions to avoid

Avoid temperatures above or at least 5 °C below flash point for any flammable liquids.

Do not heat closed containers.

Avoid contact with oxidizing agents.

10.5 Incompatible materials

Not known.

10.6 Hazardous decomposition products

Contact with water or storage under recommended conditions for one year should not produce dangerous decomposition products.

11 TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects

This mixture has not been subjected to toxicological testing as an entity. According to available data on the constituents the health classification criteria are met.

(a) acute toxicity

Calculated LD50 Oral: LD50 > 5000 mg/kg bw

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(b) skin corrosion/irritation
Skin Irritation - Cat. 2

(c) serious eye damage/irritation
Not applicable

(d) respiratory or skin sensitisation
Skin Sensitization - Cat. 1

(e) germ cell mutagenicity
Not applicable

(f) carcinogenicity
Not applicable

(g) reproductive toxicity
Not applicable

(h) STOT-single exposure
Not applicable

(i) STOT-repeated exposure
Not applicable

(j) aspiration hazard
Not applicable

12 ECOLOGICAL INFORMATION

This mixture has not been subjected to ecotoxicological testing as an entity. According to available data on the constituents the environmental classification criteria are met.

12.1 Toxicity
Environmental Hazard (acute) - Cat. 2
Environmental Hazard (chronic) - Cat. 2

12.2 Persistence and degradability
No data available

12.3 Bioaccumulative potential
No data available

12.4 Mobility in soil
No data available

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12.5 Results of PBT and vPvB assessment

Not applicable

12.6 Other adverse effects

No data available

13 DISPOSAL CONSIDERATIONS

13.1 Waste treatment methods

Product: The product should be handled according to the instructions given under sections 6, 7 and 8. Dispose of according to local or national regulations. The product should not be allowed to enter drains or the environment

Contaminated packaging: Empty packaging should be disposed according to local or national regulations by an approved waste handling

14 TRANSPORT INFORMATION

In case of accidental spillage or fire during transport, refer to instructions given under points 5, 6, 7 and 8 above.

UNO

14.1 UN-No:	3082
14.2 Proper Shipping Name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (ACETYLTETRAMETHYLOCTALINE)
14.3 Class:	9
14.4 Packing Group:	III
14.5 Environmental hazards:	Yes

Land transport (ADR/RID)

14.1 UN-No:	3082
14.2 Proper Shipping Name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (ACETYLTETRAMETHYLOCTALINE)
14.3 Class:	9
14.4 Packing Group:	III
14.5 Environmental hazards:	Yes
14.6 Special precautions for user:	Refer to section 7

Sea transport (IMDG-Code)

14.1 UN-No:	3082
14.2 Proper Shipping Name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (ACETYLTETRAMETHYLOCTALINE)
14.3 Class:	9
14.4 Packing Group:	III

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14.5 Environmental hazards: Yes
Marine pollutant: Yes
14.6 Special precautions for user: Refer to section 7
14.7 Maritime transport in bulk according to IMO instruments: Not applicable

Air transport (ICAO-IATA)

14.1 UN-No: 3082
14.2 Proper Shipping Name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (ACETYLTETRAMETHYLOCTALINE)
14.3 Class: 9
14.4 Packing Group: III
14.5 Environmental hazards: Not applicable
14.6 Special precautions for user: Refer to section 7
14.7 Maritime transport in bulk according to IMO instruments: Not applicable

15 REGULATORY INFORMATION

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture:

No relevant information available at this time.

15.2 Chemical Safety Assessment

No data available at this time.

16 OTHER INFORMATION

16.1 Revisions

March-2024 : Version 13.0 - Updates to sections 2, 3, 9, 14, 16

16.2 Key literature references

RIFM database
OECD SIDS
EU IUCLID
Supplier information

16.3 Full text of phrases used under section 2

H315 Causes skin irritation.
H317 May cause an allergic skin reaction.

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H227	Combustible liquid
H411	Toxic to aquatic life with long lasting effects.
P210	Keep away from heat/sparks/open flames/hot surfaces. No smoking.
P264	Wash thoroughly after handling.
P273	Avoid release to the environment.
P272	Contaminated work clothing should not be allowed out of the workplace.
P280	Wear protective gloves/protective clothing/eye protection/face protection.
P302+P352	IF ON SKIN: Wash with plenty of soap and water.
P333+P313	If skin irritation or rash occurs: Get medical advice/attention.
P370+P378	In case of fire: Use appropriate fire extinguishers for fire control.
P391	Collect spillage.
P362	Take off contaminated clothing and wash before reuse.
P363	Wash contaminated clothing before reuse.
P403+P235	Store in a well-ventilated place. Keep cool.

16.4 Full text of phrases used under section 3

H226	Flammable liquid and vapour.
H227	Combustible liquid
H304	May be fatal if swallowed and enters airways.
H315	Causes skin irritation.
H316	Causes mild skin irritation.
H317	May cause an allergic skin reaction.
H320	Causes eye irritation.
H331	Toxic if inhaled.
H361	Suspected of damaging fertility or the unborn child.
H400	Very toxic to aquatic life.
H401	Toxic to aquatic life.
H402	Harmful to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.
H411	Toxic to aquatic life with long lasting effects.
H412	Harmful to aquatic life with long lasting effects.

We believe that the information contained herein is current as of the date of this Safety Data Sheet. Since the use of this information and the conditions of use of the product are not within the control of Firmenich, it is the user's obligation to determine conditions of safe use of the product.

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